A 5×25 CIRCULAR FLORENTINE RECTANGLE: $F_c(25) \geq 5$, EXCEEDING THE RECORDED LOWER BOUND

DANIEL KIRTCHAKOV

Abstract. An $r \times n$ circular Florentine rectangle is an array of r rows, each a permutation of $\{0, 1, \ldots, n-1\}$ read circularly, such that for every ordered pair of distinct symbols (a, b) and every distance $m \in \{1, \ldots, n-1\}$ at most one row places b exactly m positions circularly to the right of a; $F_c(n)$ denotes the largest r for which one exists. We exhibit an explicit 5×25 circular Florentine rectangle and verify it exhaustively and exactly: its $5 \cdot 25 \cdot 24 = 3000$ ordered distance events are pairwise distinct, checked two independent ways by a standard-library program that shares no code with the search, and again by a second, independently written verifier. The object establishes $F_c(25) \geq 5$ — one more than the value 4 recorded for $n = 25$ in the *Handbook of Combinatorial Designs*, 2nd ed. (2006), Table 62.27 (p. 677), where the entry reads "4 $\cdots$ 24" and the lower bound 4 is exactly the $p-1$ of the multiplier construction (Construction 62.22(1), $p = 5$ the least prime factor of 25). The trusted base contains nothing beyond the verifier and its interpreter: there is no solver to trust, no enumeration whose completeness is assumed, and no unproven step. We do not claim priority over H.-Y. Song's 2000 paper, which we could not read; the claim is stated against the recorded Handbook value, and the array itself is new to the surveyed literature.

1. The object and the property

Fix $n \geq 2$ and the symbol set $S = \{0, 1, \ldots, n-1\}$. Read the positions of a length-n row circularly, i.e. modulo n.

Definition 1.1 (circular Florentine rectangle; [HCD, Def. 62.1]). An $r \times n$ *circular Florentine rectangle* is an $r \times n$ array in which each row is a permutation of S and which has the following property: for every ordered pair of distinct symbols (a, b) and every distance $m \in \{1, \ldots, n-1\}$, there is *at most one* row in which symbol b occupies the position m steps circularly to the right of symbol a.

Date: August 6, 2026.

2020 *Mathematics Subject Classification.* Primary 05B30; Secondary 05B15, 68V15.

Key words and phrases. Circular Florentine rectangle, Tuscan square, orthomorphism, exhaustive verification, certified computation.

Computation and authorship. The search that produced the object below, and this note, were carried out by **Claude Fable 5** (Anthropic), directed by the author, on a single Apple M4 laptop. This is a methods statement, and it is the point of the note: the object is checked by an exhaustive, exact standard-library verifier, so the provenance of the *search* is irrelevant to the validity of the *result* (see §4). For the record, the object was located by a C program (`cfr_search.c`) in the normal form $\phi_0 = \mathrm{id}$, $\phi_i(0) = 0$: the identity first row is fixed by that normal form, and the remaining four rows are a clique in the compatibility graph on the K-equivariant orthomorphisms of $\mathbb{Z}_{25}$ for the prescribed multiplier group $K = \langle 7 \rangle = \{1, 7, 18, 24\} \leq \mathbb{Z}_{25}^*$; none of that enters the verification.

Prior-art record. The primary source, chapter VI.62 ("Tuscan Squares," by W. Chu, S. W. Golomb and H.-Y. Song) of the *Handbook of Combinatorial Designs*, 2nd ed. [HCD], was read directly on August 6, 2026; the two passages the claim rests on — Definition 62.26 and the $n = 25$ entry of Table 62.27 (p. 677) — are quoted verbatim in §2. The original H.-Y. Song paper [Son00] could not be read in full (its publisher page was inaccessible on every open route tried on that date); its Table 1 was not consulted, and no priority over [Son00] is claimed (§5). A full-text novelty sweep of the same date found no surveyed source asserting $F_c(25) \geq 5$; the constructive benchmark CPro1 [CPro1] lists $(5, 25)$ as an open instance. *Draft of August 6, 2026.*

Equivalently, form for every row, every start position i, and every distance $m \in \{1, \ldots, n-1\}$ the ordered *distance event*

$$(a, b, m), \qquad a = R[i], \quad b = R[(i+m) \bmod n],$$

where R is the row. Definition 1.1 holds if and only if all such events are pairwise distinct. A single row of length n generates $n(n-1)$ events, all distinct within the row because the row is a permutation; so an $r \times n$ array has the property exactly when its $r\,n(n-1)$ events are pairwise distinct across the whole array.

	position $0, 1, \ldots, 24$ (read circularly)																								
row 0	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24
row 1	0	7	14	21	3	10	17	24	6	13	20	2	9	16	23	5	12	19	1	8	15	22	4	11	18
row 2	0	24	3	12	16	20	14	18	2	6	15	4	8	17	21	10	19	23	7	11	5	9	13	22	1
row 3	0	5	22	11	23	19	18	10	24	16	8	21	13	12	4	17	9	1	15	7	6	2	14	3	20
row 4	0	15	9	24	7	22	19	5	17	14	4	12	2	23	13	21	11	8	20	6	3	18	1	16	10

FIGURE 1. The circular Florentine rectangle CFR$(5, 25)$. Each of the five rows is a permutation of the symbol set $\{0, 1, \ldots, 24\}$; the columns are the position indices $0, 1, \ldots, 24$, read circularly. The array satisfies Definition 1.1: for every ordered pair of distinct symbols (a, b) and every distance $m \in \{1, \ldots, 24\}$, at most one row places b exactly m positions circularly to the right of a; equivalently, the $5 \cdot 25 \cdot 24 = 3000$ ordered distance events (a, b, m) are pairwise distinct. Rows 0 and 1 are the linear permutations $t \mapsto t$ and $t \mapsto 7t \pmod{25}$; rows 2, 3, 4 are non-linear. The recorded digest of the object is `sha256` = `9e2d9b33`...`779ffdcc`.

2. The claim

What F_c records. The primary source defines the quantity our object bounds. Quoting [HCD, Def. 62.26, p. 677] verbatim:

> "Let $F_c(n)$ denote the maximum integer F_c such that an $F_c \times n$ circular florentine rectangle exists."

The basic bounds recorded there are $p - 1 \leq F_c(n) \leq n - 1$, where p is the least prime factor of n: the lower bound is the multiplier construction [HCD, Construction 62.22(1), p. 676], which for the least prime factor p of n takes the rows $t \mapsto ct \pmod{n}$ over a suitable set of multipliers c and yields a $(p-1) \times n$ circular Florentine rectangle; the upper bound $n - 1$ is immediate. For $n = 25$ the least prime factor is $p = 5$, so the construction gives 4 rows, and [HCD, Table 62.27, p. 677] — headed "Possible values of $F_c(n)$ and $F(n)$" — records the $n = 25$ entry, verbatim, as

$$25 \qquad 4 \ \cdots \ 24,$$

i.e. a best recorded lower bound of 4 (the value $p - 1$) and the trivial upper bound $24 = n - 1$; the value of $F_c(25)$ within that range is left undetermined.

The result.

Theorem 2.1. *The array of Figure 1 is a circular Florentine rectangle. Hence*

$$\boxed{F_c(25) \ \geq \ 5}$$

which is one more than the value 4 recorded for $n = 25$ in the Handbook of Combinatorial Designs, 2nd ed. (2006), Table 62.27 (p. 677).

Proof. By Definition 1.1 it suffices that the $5 \cdot 25 \cdot 24 = 3000$ ordered distance events of the five rows in Figure 1 are pairwise distinct; this is confirmed exhaustively in §3. The array is then a 5×25 circular Florentine rectangle, so by [HCD, Def. 62.26] the maximum r for which an $r \times 25$ such rectangle exists is at least 5. □

The two linear rows $t \mapsto t$ and $t \mapsto 7t$ are multiplier rows of the form $t \mapsto ct$ that underlies the multiplier construction [HCD, 62.22(1)]; the remaining three rows are non-linear (this is confirmed by the second verifier `verify_cfr.py` of §3, which finds no multiplier c with $R[t] \equiv ct$ for rows 2, 3, 4; the shipped `verify_cfr525.py` performs no linearity test). The recorded lower bound $4 = p - 1$ is exactly what the multiplier construction alone delivers; Figure 1 exceeds it by adjoining a fifth, non-linear row that remains event-disjoint from the other four.

3. Verification

The verification is *exhaustive* and *exact*: it enumerates all 3000 ordered distance events directly, in integer arithmetic, and confirms they are distinct. There is no sampling, no floating point, and no randomized or heuristic step.

The shipped verifier. `verify_cfr525.py` (173 lines, Python standard library only — `json`, `hashlib`, `sys`, `os` — with no POSIX-only call, so it runs unchanged on any platform with CPython 3) performs, against the object file `CFR_5_25.json`:

(1) *SHA-256.* It recomputes the digest of the object file and requires it to equal the recorded value `9e2d9b33…779ffdcc`; and it cross-checks the rows read from disk against a second, independent in-source transcription of the same data.
(2) *Shape and permutation.* It checks $r = 5$, $n = 25$, and that each row is a permutation of $\{0, \ldots, 24\}$.
(3) *Property, method A (literal circular window).* For every row, every start position i, and every distance $m \in \{1, \ldots, 24\}$ it forms the event (a, b, m) with $a = R[i]$, $b = R[(i + m) \bmod 25]$, and counts the distinct events. The property holds iff the count equals $r\,n(n - 1) = 3000$.
(4) *Property, method B (per ordered pair via positions).* For each ordered pair (a, b) of distinct symbols it computes, in each row, the unique distance $(\mathrm{pos}[b] - \mathrm{pos}[a]) \bmod 25$, and requires these to be pairwise distinct across the rows. It then requires methods A and B to agree.

Run on the shipped object today, the verifier reports, in its final lines (the long verdict line soft-wrapped here to fit the text block),

```
[sha256] match    = True
[shape ] r=5, n=25, 5 rows of length n: True
[cross ] disk rows == independent transcription: True
[perm  ] every row is a permutation of {0..24}: True
[meth A] window events 3000/3000 distinct (expected 3000): True
[meth B] per-pair distances 3000/3000 distinct (expected 3000): True
[cross ] methods A and B agree: True

VERIFIED: the object is a valid CFR(5,25); all 3000 ordered distance
events are distinct, so F_c(25) >= 5.
```

As a negative control, swapping any two symbols within a row (which breaks the digest and the property) is rejected: the event count falls below 3000 and the run exits non-zero.

A second, independent verifier. A second verifier, `verify_cfr.py` (in the program's `construct/florentine` directory), was written separately from the first and from the search. It re-checks the same object through unrelated code — the SHA-256, the permutation property, both event-counting methods, and, additionally, the $K = \langle 7 \rangle$ multiplier equivariance and the per-row linearity classification — and returns the same verdict (`OVERALL: OBJECT IS A VALID CFR(5,25): True`). The two verifiers share no code and agree.

4. The trusted base

What a skeptic must believe to accept Theorem 2.1 is unusually small, and it is worth stating exactly, because it is smaller than in the program's earlier certified notes (each of which declared at least one trust assumption, such as the completeness of an external enumerator).

Machine-checked, from raw data: that the object file has the recorded SHA-256; that each row is a permutation of $\{0, \ldots, 24\}$; and that all 3000 ordered distance events are pairwise distinct, established two independent ways and reproduced by a second, independently written verifier.

Assumed: only that CPython and the operating system execute the 173-line `verify_cfr525.py` as written, and that its transcription of Definition 1.1 ([HCD, Def. 62.1/62.26]) into code is faithful — which a reader confirms by reading the verifier against the definition. There is nothing else. There is no external solver, no enumeration whose completeness must be trusted, no unproven lemma, and no floating-point or randomized computation. The check is a finite, exact, exhaustive enumeration of 3000 events, and it either finds all of them distinct or it does not.

In particular, the search apparatus — `cfr_search.c`, the multiplier group K, the clique computation — is *not* in the trusted base. The object stands or falls on its own, independently of how it was found.

5. Novelty, and what is not claimed

The improvement is over the recorded value. Theorem 2.1 asserts $F_c(25) \geq 5$ and states the improvement against the value 4 recorded for $n = 25$ in Table 62.27 (p. 677) of [HCD] — a citable secondary source whose $n = 25$ entry reads "4 $\cdots$ 24" and which cites, for that table, references predating the specific determination of $F_c(25)$. A full-text novelty sweep on August 6, 2026 found no surveyed source asserting $F_c(25) \geq 5$: the recent literature that uses circular Florentine rectangles quotes only the basic bounds $p - 1 \leq F_c(N) \leq N - 1$, and the constructive benchmark CPro1 [CPro1] lists $(5, 25)$ among its open instances, with a property definition matching Definition 1.1. The array of Figure 1 is new to the surveyed literature.

No priority over Song 2000 is claimed. The one paper that might already contain a determination of $F_c(25)$ is H.-Y. Song, *The existence of circular Florentine arrays* [Son00]. Its publisher page was inaccessible on every open route tried on August 6, 2026, so its Table 1 was not consulted. We therefore make *no* claim of priority over [Son00]. The claim of Theorem 2.1 is deliberately anchored on the recorded Handbook value of 4, not on the assertion that 4 was the best value obtainable anywhere: we state that the object exceeds the recorded lower bound, and no more.

6. Artifacts

Two files are shipped with this note; both are pinned by SHA-256.

file	bytes	SHA-256
`CFR_5_25.json`	1236	`9e2d9b33…779ffdcc`
`verify_cfr525.py`	—	`9d7363d0…37dd0bb`

The full digests are

```
CFR_5_25.json    9e2d9b339a79d13783d2e24efde49e5ed7904418f2179bf7d3f0143d779ffdcc
verify_cfr525.py  9d7363d08ce2cd90b194ce1f8c318545501006c14b3ea1cb7871586ec37dd0bb
```

To reproduce the verification on any machine with CPython 3, from the directory holding the two files:

```
python3 verify_cfr525.py          # reads CFR_5_25.json beside the script
python3 verify_cfr525.py PATH     # or an explicit path to the object
```

No compiler, solver, or third-party package is required; the script imports only the Python standard library and terminates in well under a second.

Acknowledgments

The definitions, the quantity $F_c(n)$, the multiplier construction that fixes the recorded lower bound 4, and the value tables are those of the *Handbook of Combinatorial Designs* chapter on Tuscan squares by Wai Chu, Solomon W. Golomb and Hong-Yeop Song [HCD]. The $(5, 25)$ instance is posed as open by the CPro1 benchmark [CPro1]. The computation and drafting were AI-assisted as stated in the first footnote; the object is checked by an exhaustive standard-library verifier, so that assistance bears on how the object was found, not on whether it is valid.

References

[HCD] C. J. Colbourn and J. H. Dinitz (eds.), *Handbook of Combinatorial Designs*, 2nd ed., CRC Press, Boca Raton, 2006. Chapter VI.62, "Tuscan Squares," by W. Chu, S. W. Golomb and H.-Y. Song, pp. 673–678. Read directly on August 6, 2026; the claim rests on Definition 62.26 and the $n = 25$ entry of Table 62.27 (p. 677), "4 $\cdots$ 24," with the lower bound $4 = p - 1$ from Construction 62.22(1) ($p = 5$).

[Son00] H.-Y. Song, *The existence of circular Florentine arrays*, Comput. Math. Appl. **39** (2000), no. 11, 31–35. DOI 10.1016/S0898-1221(00)00104-8. Not read in full (publisher page inaccessible on every open route tried on August 6, 2026); its Table 1 was not consulted, and no priority over this paper is claimed.

[CPro1] Constructive-Codes, *CPro1* benchmark suite, constructive problem `circular-florentine-rectangle`, which lists the open instances $(6, 21)$, $(5, 25)$, $(5, 27)$, $(4, 33)$ with the property of Definition 1.1. On-disk copy consulted August 6, 2026.

Independent researcher (05oz); no institutional affiliation.
Email address: `daniel@halfounce.io`
URL: `https://halfounce.io`